Kavya Patel

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PROFILE

Computer Science and Engineering student (Data Science specialization) building end-to-end AI and data solutions, from multimodal deep learning and computer vision to full-stack applications with REST APIs. Completed a software development internship and built projects in healthcare AI, video surveillance, and retail analytics. Seeking Data Science, Machine Learning, or Software Engineering roles.

EDUCATION

Vellore Institute of Technology, Chennai, India **2023 – 2027**

B.Tech in Computer Science and Engineering (Specialization in Data Science) CGPA: 8.26/10

SKILLS

- **Languages:** Python, Java, C++, C, JavaScript, SQL
- **Data Science & AI:** Machine Learning, Deep Learning, Natural Language Processing (NLP), Computer Vision, Generative AI, EDA, Scikit-learn, TensorFlow, Keras, PyTorch, XGBoost, OpenCV, LangChain, LangGraph, MCP
- **Analytics, DevOps & Tools:** Pandas, NumPy, Power BI, KNIME, R
- **Web Technologies:** HTML5, CSS3, React.js, Node.js, Express.js, Flask, REST APIs, MongoDB, MySQL
- **Core Concepts:** Data Structures and Algorithms, OOP, DBMS, Operating Systems, Computer Networks

EXPERIENCE

Sahajanand Laser Technology Ltd. (SLTL Group) **May 2026 – June 2026**

Software Development Intern **Gandhinagar, India**

- Developed a full-stack Online Commerce and Inventory Management System to automate product management, inventory tracking, and order processing workflows, streamlining day-to-day business operations.
- Implemented secure user authentication and role-based access control (RBAC) to restrict access by user role, and built order management workflows that improved application security and reliability.
- Built backend services for inventory and product management, optimizing database operations and REST API performance to support efficient, dependable data handling.
- Designed responsive, user-friendly interfaces and collaborated with the development team using Git version control and industry-standard practices.

PROJECTS

DERMA-GUARD: Agentic Multimodal Skin Lesion Assessment System GitHub

- Engineered an AI dermatology assistant that knows when not to trust itself: 7 multimodal models (ViT, Residual MLP, Bio_ClinicalBERT) with adaptive gated fusion over photos, patient data, and clinical notes.
- Built an evidence manager scoring quality, reliability, and cross-modal conflict to accept, acquire more evidence, or escalate to a dermatologist, backed by conformal prediction sets.
- Orchestrated Vision, Scribe, Librarian, and Devil's Advocate agents with RAG and a local Qwen2.5 LLM with hallucination checks shipped with RestAPI Docker, and 15-scenario stress tests.
- **Stack:** *Python, PyTorch, Transformers, RestAPI, RAG, LLM Agents, Bio_ClinicalBERT, ViT, Scikit-learn, Docker*

Automated Suspicious Activity Detection GitHub

- Developed an intelligent video surveillance system using CNN-LSTM and YOLOv8 for real-time suspicious activity detection and object localization from CCTV video streams.
- Implemented video preprocessing, HOG and ORB feature extraction, object detection, temporal activity recognition, and model optimization using TensorFlow, Keras, and OpenCV.
- Designed an end-to-end deep learning pipeline for activity classification and real-time inference.
- **Stack:** *Python, TensorFlow, Keras, OpenCV, YOLOv8, CNN-LSTM, HOG, ORB, Computer Vision*

Customer Intelligence Sales Prediction Platform GitHub

- Created a retail analytics platform with Python, Node.js, and MongoDB for sales analysis, customer behavior studies, and product performance.
- Conducted data preprocessing, feature engineering, exploratory data analysis (EDA), and statistical testing; trained machine learning models for sales forecasting and customer segmentation.
- Integrated NLP-based sentiment analysis and interactive visualization dashboards to deliver actionable insights for data-driven decisions.
- **Stack:** *Python, Pandas, NumPy, Scikit-learn, NLP, Node.js, MongoDB, Data Visualization*

ACHIEVEMENTS & CERTIFICATIONS

- Generative AI Using IBM Watson – IBM Career Education Program